

|                        | #q | #G   | This Work |      |             | CAV'23 |       |              |
|------------------------|----|------|-----------|------|-------------|--------|-------|--------------|
|                        |    |      | trans     | ver  | total       | trans  | ver   | total        |
| BV-ALL                 | 17 | 27   | 0.0s      | 0.0s | <b>0.0s</b> | 1.4s   | 0.9s  | <b>2.3s</b>  |
|                        | 19 | 30   | 0.0s      | 0.0s | <b>0.0s</b> | 6.4s   | 4.4s  | <b>10.8s</b> |
|                        | 21 | 33   | 0.0s      | 0.0s | <b>0.0s</b> | 25s    | 20.2s | <b>45.2s</b> |
|                        | 23 | 36   | 0.0s      | 0.0s | <b>0.0s</b> | 2m21s  | 1m31s | <b>3m52s</b> |
|                        | 25 | 39   | 0.0s      | 0.0s | <b>0.0s</b> | TO     |       |              |
| GHZ-ALL                | 8  | 8    | 0.0s      | 0.0s | <b>0.0s</b> | 0.6s   | 0.2s  | <b>0.8s</b>  |
|                        | 9  | 9    | 0.0s      | 0.0s | <b>0.0s</b> | 2.7s   | 1.1s  | <b>3.8s</b>  |
|                        | 10 | 10   | 0.0s      | 0.0s | <b>0.0s</b> | 11s    | 4.8s  | <b>15.8s</b> |
|                        | 11 | 11   | 0.0s      | 0.0s | <b>0.0s</b> | 45s    | 20.1s | <b>1m5s</b>  |
|                        | 12 | 12   | 0.0s      | 0.0s | <b>0.0s</b> | TO     |       |              |
| GROVER-ALL             | 20 | 544  | 0.0s      | 0.2s | <b>0.2s</b> | 0.5s   | 3s    | <b>3.5s</b>  |
|                        | 23 | 927  | 0.0s      | 0.5s | <b>0.5s</b> | 1.9s   | 10.5s | <b>12.4s</b> |
|                        | 26 | 1475 | 0.0s      | 1s   | <b>1s</b>   | 7.5s   | 43s   | <b>50.5s</b> |
|                        | 29 | 2408 | 0.0s      | 1.9s | <b>1.9s</b> | 30s    | 3m39s | <b>4m9s</b>  |
|                        | 32 | 3711 | 0.0s      | 3.6s | <b>3.6s</b> | TO     |       |              |
| GROVER-ITER            | 20 | 79   | 0.0s      | 0.1s | <b>0.1s</b> | 0.8s   | 2.9s  | <b>3.7s</b>  |
|                        | 23 | 91   | 0.0s      | 0.1s | <b>0.1s</b> | 3.0s   | 8.4s  | <b>11.4s</b> |
|                        | 26 | 103  | 0.0s      | 0.1s | <b>0.1s</b> | 12s    | 31.7s | <b>43.7s</b> |
|                        | 29 | 115  | 0.0s      | 0.1s | <b>0.1s</b> | 51s    | 2m23s | <b>3m14s</b> |
|                        | 32 | 127  | 0.0s      | 0.2s | <b>0.2s</b> | TO     |       |              |
| MCTOFFOLI <sup>1</sup> | 16 | 15   | 0.0s      | 0.0s | <b>0.0s</b> | 1.2s   | 0.4s  | <b>1.6s</b>  |
|                        | 18 | 17   | 0.0s      | 0.0s | <b>0.0s</b> | 4.6s   | 1.7s  | <b>6.3s</b>  |
|                        | 20 | 19   | 0.0s      | 0.0s | <b>0.0s</b> | 18.2s  | 7.5s  | <b>25.7s</b> |
|                        | 22 | 21   | 0.0s      | 0.0s | <b>0.0s</b> | 1m14s  | 30.4s | <b>1m44s</b> |
|                        | 24 | 23   | 0.0s      | 0.0s | <b>0.0s</b> | TO     |       |              |